

PathIQ Navigators: MVP build

Reporting period: 15 to 26 September 2026 · Date: 26 September 2026

Summary

The Nairobi MVP of the PathIQ driver app is **built end to end and working against the live backend**. A driver can open the app with no sign-up, search for a destination, compare scored route options, and drive with turn-by-turn voice navigation. Along the way they get Hidden Gem alerts, and each trip counts towards their IQ Score. Scouts can submit road reports and gems with photos, and admins review them before anything goes live.

Almost everything left before a public pilot is a **decision, account or piece of content on your side**, not development work. Section 5 lists these. The largest is replacing the demo Google Maps key with your own restricted keys. The other is gathering 50 to 100 real, verified Hidden Gems.

13 of 14 MVP feature areas built and tested	61 / 61 automated security checks passing	5 app screens: Home, Map, History, Gems, Profile	8 marketing and legal pages
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1. Milestones this period

Date	Milestone
15 Sep	Starting point: front-end prototype with a landing page and app screens on sample data. No map, backend or real sign-in.
20 Sep	Product description and MVP scope agreed. Backend set up: database, review workflow for Scout submissions, and the route-scoring engine.
23 Sep	Guest mode: drivers use the app straight away, with no login wall. Nearby gems and road reports served from the database.
25 Sep	Google Maps, turn-by-turn navigation, gem alerts, trip recording and IQ Score, Scout photos, installable app and security hardening.
25 Sep	Map redesign in PathIQ's own style. Route planning moved onto the map. Written directions, live traffic shading, direction arrows, turn icons and one icon set for gems and reports.

2. What has been delivered

Map

- A live Google map in PathIQ's own colours. Google's own shop and bus-stop pins are hidden, so PathIQ's gems and road reports stand out.
- A layer switcher for Hidden Gems, road reports and live traffic. Dark and light modes.
- Tap anywhere to drop a pin and get a route there. Scouts can also report a road problem or add a gem at that spot.
- Search shows recent destinations and saved gems. Typing searches Google Places in Kenya alongside PathIQ gems.

Route intelligence

- Google's alternative routes are scored against PathIQ's verified road reports and labelled **Recommended**, **Fastest** and **Best Road**.
- A panel on the map compares every option side by side: time, distance, road and traffic scores, and the number of reported problems.
- Routes are drawn like Google Maps: live traffic shading in amber and red, direction arrows and turn icons. Reported problems are shown as dotted stretches in their severity colour.
- Stale reports fade out on their own. For example, an accident stops affecting routes after about 20 hours.

Turn-by-turn navigation

- Voice and on-screen directions from the driver's live GPS position, snapped to the route.
- Automatic rerouting when the driver leaves the route, and arrival detection.
- A full written list of steps, plus a "Then..." hint when two turns come close together.
- The screen stays on during a trip so tracking and alerts keep working.

Hidden Gems

- Browse and filter by category (Attractions, Hotels, Food, Scenic, Fuel, Facilities), then save and rate gems.
- While driving: a "Hidden gem ahead" alert within 600 m, and a "You just passed" alert with a Save button.
- Alerts are voiced and shown once per gem per trip. Drivers can filter them by category and by how far they are willing to detour.

Trips, History and IQ Score

- Each trip is recorded as a summary. No GPS trail is stored, which keeps the privacy footprint small.
- The History tab lists past trips by day with the outcome, distance, time, road score and gems passed. From there the driver can **Go again** or **Remove** a trip.
- IQ Score v1 covers the last 30 days: 50% road quality of the routes chosen, 30% trips completed and 20% staying on route. Home and Profile show real "Trips this month" and "Gems found" counts.

Scouts and admin

- Any user can become a Scout from their Profile and submit road reports or gems, with GPS and an optional photo.
- Admins review the queue with a photo preview, then approve or reject. Only approved items appear to drivers.
- Payouts are recorded by hand for now, as planned for the MVP.

Installable app, website and legal

- Installs to a phone's home screen like a native app and opens with no signal.
- Marketing site: Home, Map, Hidden Gems, Scout Program, Developers and Pricing, with a mobile menu.
- First drafts of the Privacy and Terms pages, written to match what the app actually does.

Security

- Every database table is protected by access rules. Drivers can only see and change their own data, and the IQ Score cannot be inflated by editing trip records.
- Rate limits on route requests, trip starts, submissions and photo uploads. These include an overall cap on Google usage to protect the bill.
- An automated suite of 61 security checks runs against the live backend. All 61 pass.

3. MVP scorecard

Measured against the agreed MVP scope:

Area	Status	Notes
Sign-up, sign-in and guest mode	Built	No login wall. Guests can upgrade to a full account.
Live map	Built	Gems, road reports and traffic layers.
Route options and scoring	Built	Recommended, Fastest and Best Road.
Turn-by-turn navigation	Built	Voice, rerouting, arrival and written steps.
Gem alerts (ahead and just passed)	Built	Once per trip, with filters.
Gems: browse, filter, save, rate	Built	
Road-condition layer	Built	Scout-reported and admin-reviewed.
Home dashboard and Profile	Built	Real stats and a working notifications toggle.
Driver IQ Score	Built (v1)	Simple formula, as agreed for the MVP.
Scout submissions and review queue	Built	Manual payouts, as planned.
Installable app and offline start	Built	
Security and rate limits	Built	61 of 61 checks pass.
Privacy and Terms	Drafted	Need legal review and a contact address.
Seed data: 50 to 100 verified gems	Not done	11 gems so far. Needs real, visited places.

4. Testing and quality

- Driving was tested end to end in the browser using simulated GPS: route planning, navigation, rerouting, arrival, gem alerts and trip recording.
- The security suite (61 checks) and the route-scoring unit tests pass against the live backend.
- The production build and code checks (lint) pass as of today.
- Installability and offline start were confirmed in Chrome. Real-device testing on Android phones is the next step.

5. What we need from you before launch

These items block a public pilot. None of them needs development work, but each one needs an account, a decision or content that only your team can provide.

#	Item	What's needed
1	Google Maps keys	The app currently runs on a single demo key. The owner of your Google Cloud account should create a restricted browser key (Maps JavaScript and Places, locked to your domain) and a server key (Routes only). They should also set daily quotas and a budget alert. We will then switch the app over, and the demo key should be deleted.
2	Email provider	Email confirmation on sign-up needs a custom email (SMTP) provider. Please pick one, for example your domain's mail service, Resend or SendGrid.

#	Item	What's needed
3	Legal review	A lawyer should review the Privacy and Terms drafts against Kenya's Data Protection Act 2019. We also need the contact address to publish on them.
4	Verified Hidden Gems	The MVP target is 50 to 100 real, visited places in Nairobi. There are 11 today. This is the first job for the Scout programme.
5	Landing-page figures	The homepage stats (for example 42K+ drivers and 1,284 Scouts) and the gem photos are placeholders. Please supply real figures and photos, or approve removing them.
6	Traffic-shading cost	Google bills live traffic on routes at a higher rate. It is on now and can be switched off with one setting. Please confirm whether to keep it for the pilot.

6. Known limitations

- **Phones in the background:** browsers pause GPS when the screen is off, especially on iPhone. Drivers need to keep the app open during a trip. A native app, planned for 2027, removes this limit.
- **Few road reports on routes yet:** the sample reports are not placed on actual roads, so routes rarely show problems until Scouts submit real ones. The feature itself has been verified with a test report.
- **Trip results are reported by the phone:** arrival and route choice are not yet checked on the server. This is fine for a pilot, but should be hardened before the IQ Score carries any rewards.
- **Placeholder links:** the About, Blog, Careers, Docs and Scout Portal links in the website footer are not built yet.

7. Proposed next steps

When	Work
Now	Switch to your Google keys and email provider as soon as they are available. Publish the reviewed legal pages. Test on real Android phones in Nairobi traffic.
Pilot	Scouts gather verified gems and road reports. Replace the placeholder figures. Run a small closed pilot with real drivers.
Polish	Heading-up map during navigation, greying out the part of the route already driven, and spoken warnings for reported road problems ahead.
Next phase	Design the business/partner and developer (API) account types. Scout portal with earnings and M-Pesa payouts. Decide the free/paid split for drivers.

Questions or feedback on this report are welcome. We are happy to walk you through the app live.